

# Policy Iteration Example

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Consider the grid

|     |     |
|-----|-----|
| $A$ | $B$ |
| $C$ | $G$ |

where  $G$  is the terminal goal state.

At each non-terminal state, the agent may choose one of the actions Up, Down, Left, or Right. If an action would move the agent outside the grid, the agent stays in the same state. Every move that does not enter the goal gives reward  $-1$ . A move into the goal state  $G$  gives reward 0 and the episode terminates. The discount factor is fixed at  $\gamma = 0.9$ . We will apply Policy Iteration.

We start with a random initial policy

$$\pi_0(A) = \text{Up}, \quad \pi_0(B) = \text{Up}, \quad \pi_0(C) = \text{Left}.$$

Under this policy, the agent loops forever at each of the three non-terminal states, because Up from  $A$  leaves the agent at  $A$ , Up from  $B$  leaves the agent at  $B$ , and Left from  $C$  leaves the agent at  $C$ .

(i) **Policy Evaluation** We first evaluate this policy. At state  $A$ , the Bellman equation is

$$V^{\pi_0}(A) = -1 + 0.9V^{\pi_0}(A).$$

Solving gives

$$V^{\pi_0}(A) = -10.$$

By the same reasoning,

$$V^{\pi_0}(B) = -10 \quad \text{and} \quad V^{\pi_0}(C) = -10.$$

Since  $G$  is terminal,

$$V^{\pi_0}(G) = 0.$$

So after policy evaluation, the values are

$$V^{\pi_0}(A) = -10, \quad V^{\pi_0}(B) = -10, \quad V^{\pi_0}(C) = -10, \quad V^{\pi_0}(G) = 0.$$

(ii) **Policy Improvement** We now improve the policy by choosing, at each state, the action that maximizes the one-step look-ahead value.

At state  $A$ , every action gives the same value:

$$-1 + 0.9(-10) = -10.$$

So there is a tie, and we keep the current action Up.

At state  $B$ , moving Down enters the goal, so its value is

$$0.$$

Every other action gives

$$-1 + 0.9(-10) = -10.$$

Therefore the improved action at  $B$  is Down.

At state  $C$ , moving Right enters the goal, so its value is

$$0.$$

Every other action gives

$$-1 + 0.9(-10) = -10.$$

Therefore the improved action at  $C$  is Right.

The improved policy is thus

$$\pi_1(A) = \text{Up}, \quad \pi_1(B) = \text{Down}, \quad \pi_1(C) = \text{Right}.$$

We now evaluate this new policy. Since  $B$  goes directly to the goal, we have

$$V^{\pi_1}(B) = 0.$$

Since  $C$  also goes directly to the goal, we have

$$V^{\pi_1}(C) = 0.$$

At state  $A$ , the policy still loops upward, so

$$V^{\pi_1}(A) = -1 + 0.9V^{\pi_1}(A),$$

which again gives

$$V^{\pi_1}(A) = -10.$$

Also,

$$V^{\pi_1}(G) = 0.$$

So after this evaluation,

$$V^{\pi_1}(A) = -10, \quad V^{\pi_1}(B) = 0, \quad V^{\pi_1}(C) = 0, \quad V^{\pi_1}(G) = 0.$$

We improve the policy again. At state  $A$ , moving Right goes to  $B$ , whose value is now 0, so

$$Q(A, \text{Right}) = -1 + 0.9(0) = -1.$$

Moving Down goes to  $C$ , whose value is also 0, so

$$Q(A, \text{Down}) = -1 + 0.9(0) = -1.$$

By contrast, moving Up or Left keeps the agent in  $A$ , so each of those gives

$$-1 + 0.9(-10) = -10.$$

Therefore the best actions at  $A$  are Right and Down. We choose Right at random.

The new policy is

$$\pi_2(A) = \text{Right}, \quad \pi_2(B) = \text{Down}, \quad \pi_2(C) = \text{Right}.$$

We now evaluate this policy. Since  $B$  goes directly to  $G$ ,

$$V^{\pi_2}(B) = 0.$$

Since  $C$  goes directly to  $G$ ,

$$V^{\pi_2}(C) = 0.$$

At  $A$ , the policy moves to  $B$ , so

$$V^{\pi_2}(A) = -1 + 0.9V^{\pi_2}(B) = -1 + 0.9(0) = -1.$$

And again,

$$V^{\pi_2}(G) = 0.$$

Thus

$$V^{\pi_2}(A) = -1, \quad V^{\pi_2}(B) = 0, \quad V^{\pi_2}(C) = 0, \quad V^{\pi_2}(G) = 0.$$

A final policy improvement check shows that at  $A$ , both Right and Down give value  $-1$ , while Up and Left are worse. At  $B$ , Down is still optimal. At  $C$ , Right is still optimal. Therefore the policy does not change, and policy iteration terminates.

The final policy is

$$\pi^*(A) = \text{Right}, \quad \pi^*(B) = \text{Down}, \quad \pi^*(C) = \text{Right}.$$

An equally good optimal choice at  $A$  would be Down instead of Right.